Statistics Formula Sheet — High School / Basic Level

Descriptive Statistics, Probability, Hypothesis Testing Basics

Topic	Formula / Concept	Explanation / Example
Mean	x_bar = (1/n) * Sum(x_i)	Arithmetic mean. Example: (2+4+6)/3 = 4
Median	Sorted order, middle value	n=7: 4th value. n=6: mean of middle two
Mode	Most frequent value	In [1,1,2,3,3,3,4] the mode is 3
Range	R = max - min	In [2,5,8,12]: R = 12-2 = 10
Variance	s^2 = (1/(n-1)) * Sum(x_i - x_bar)^2	Mean squared deviation from mean
Std. Deviation	s = sqrt(s^2)	Square root of variance. Same unit as data
Coeff. of Variation	CV = s / x_bar * 100	Relative spread in %. Ratio scale only
Standardization	z = (x - x_bar) / s	z-score: deviation in SD units. Mean 0, SD 1
Correlation (r)	r = cov(x,y) / (s_x * s_y)	Range: -1 to +1. 0 = no linear relationship
Proportion	p = k / n	Proportion of k successes in n trials
Probability	P(A) = |A| / |Omega|	Laplace: favorable / possible outcomes
Conditional Prob.	P(A|B) = P(A and B) / P(B)	Probability of A given B
Independence	P(A and B) = P(A) * P(B)	If A and B are independent
Expected Value	E(X) = Sum(x_i * p_i)	Mean of the random variable
Binomial Dist.	P(X=k) = C(n,k) * p^k * (1-p)^(n-k)	n trials, success probability p
Normal Dist.	phi(z) = 1/sqrt(2*pi) * e^(-z^2/2)	Bell curve. Standard normal N(0,1)
68-95-99.7 Rule	mu +/- 1s: 68%; +/- 2s: 95%; +/- 3s: 99.7%	Rule of thumb for normal distribution
CI (Mean)	CI = x_bar +/- z * s / sqrt(n)	e.g. z=1.96 for 95% CI. Larger n = narrower CI
Significance Level	alpha = 0.05 (5%)	Probability of Type I error
p-value	p < alpha -> reject H0	Probability of observed result under H0
Type I Error	Reject H0 when H0 is true	False positive. Risk = alpha
Type II Error	Retain H0 when H1 is true	False negative. Risk = beta